POLICY BRIEF #47

Action Points for the Ministry of Food Processing Industries (MoFPI)

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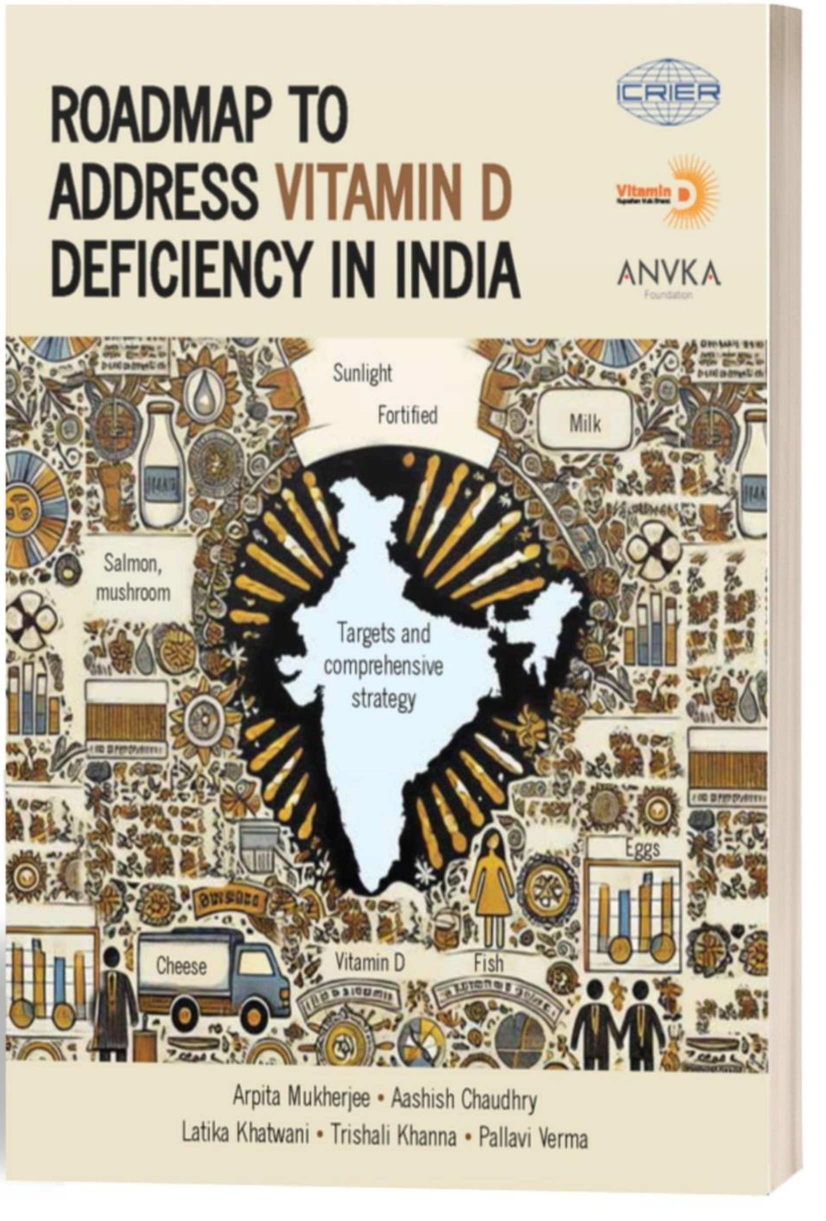

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Abstract

As India moves toward achieving nutrition security, micronutrient deficiency remains a pressing public health concern – most notably Vitamin D deficiency, which now affects one in every five Indians. The ICRIER-ANVKA Foundation 2025 report, titled, *“Roadmap to Address Vitamin D Deficiency in India”*, highlights the potential of food and diet in addressing Vitamin D deficiency. Despite being one of the world’s largest producers of Vitamin D-rich foods such as milk, eggs and fish, such foods are not reaching the population at scale. The Food Safety and Standards Authority of India (FSSAI) permits the voluntary fortification of only milk and edible oil with Vitamin D. The National Institute of Food Technology, Entrepreneurship and Management (NIFTEM), under the Ministry of Food Processing Industries (MoFPI), is actively engaged in research on food-based Vitamin D fortification.

International organisations like the World Health Organization (WHO), the Food and Agriculture Organization (FAO), and the United Nations International Children's Fund (UNICEF) have recommended food fortification as one of the most cost-effective strategies to combat micronutrient deficiencies. They have also recommended the fortification of food with multiple micronutrients. Countries such as Jordan, Nigeria, Sweden, Canada, the USA and Finland have fortified many food products including milk, cheese, fat spreads, yogurt, wheat, cereals, biscuits, juice, etc., with Vitamin D.

This policy brief outlines five action points that the MoFPI may undertake to address Vitamin D deficiency through food fortification and to support processing of food to retain Vitamin D. These range from expanding fortification to staples, incentivising research and production, launching targeted pilot programmes through multi-stakeholder partnerships, promoting the production and consumption of fortified foods and supporting micro, small and medium enterprises (MSMEs) to produce foods rich in micronutrients through subsidies and capacity building initiatives. By embedding nutrition in food processing, MoFPI can play a transformative role in realising a *"Vitamin D Kuposhan Mukt Bharat”.*

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List of Abbreviations

CDSCO	Central Drugs Standard Control Organisation
CFTRI	Central Food Technological Research Institute
EU	European Union
FAO	Food and Agriculture Organization
FSSAI	Food Safety and Standards Authority of India
GAIN	Global Alliance for Improved Nutrition
HRQL	Health-related quality of life
ICER	Incremental Cost Effectiveness Ratio
ICMR-NIN	Indian Council of Medical Research – National Institute of Nutrition
IP	Intellectual Property
ISBMR	Indian Society for Bone and Mineral Research
MoFPI	Ministry of Food Processing Industries
MoHFW	Ministry of Health and Family Welfare
MoWCD	Ministry of Women and Child Development
MSME	Micro, Small and Medium Enterprises
NDDB	National Dairy Development Board
NIFTEM	National Institute of Food Technology, Entrepreneurship and Management
NPPA	National Pharmaceutical Pricing Authority
PHFI	Public Health Foundation of India
QALY	Quality Adjusted Life Year
R&D	Research and Development
SAMPADA	Survey for Assessment of Markers of Population Health Activity, Diet and Anthropometry
UNICEF	United Nations International Children's Fund
UNWFP	United Nations World Food Programme
USA	United States of America
UT	Union Territory
UV	Ultraviolet
WHO	World Health Organization

Action Points for the Ministry of Food Processing Industries (MoFPI)

Arpita Mukherjee, Aashish Chaudhry, Trishali Khanna, Latika Khatwani and Pallavi Verma

1. Fortifying India's Future: Tackling Vitamin D Deficiency through Food-Based Solutions

Micronutrient deficiency is a pressing public health challenge in India, which accounts for nearly half of the world's micronutrient-deficient population. Although India has made significant strides in addressing some micronutrient deficiencies, such as iron and iodine deficiency through double fortified salt, Vitamin D[1] deficiency, which one in five Indians suffer from, now demands urgent and focused attention. This deficiency poses lifelong health risks with far-reaching consequences, including rickets in children, osteomalacia and osteoporosis in adults, and poor foetal bone development during pregnancy.[2]

India is well-equipped to tackle this deficiency through food. The country is the world's largest producer of Vitamin D-rich foods like milk, and the second-largest producer of eggs and fish (as of 2024).[3] While these are consumed by many Indians, other foods rich in Vitamin D like salmon, cod liver oil and mushrooms are only consumed by a niche group of the population and are not part of the regular Indian diet. Moreover, around 30% of the Indian population is vegetarian,[4] while many Vitamin D-rich foods are animal-based.

Sources of Vitamin D

Sunshine: Vitamin D, also known as the 'sunshine vitamin,' is primarily obtained through exposure to sunlight.

Foods: Fish, eggs, mushrooms, cheese, etc., fortified products like milk, margarine, yogurt, juices and cereals.

Vitamin D Supplements: Vitamin D_2 (Ergocalciferol, mostly plant-based) and D_3 (Cholecalciferol, mostly animal-based).

2. Food Fortification: An Approach to Address Vitamin D Deficiency

Food fortification[5] has been proven to be an effective strategy to address multiple micronutrient deficiencies, including Vitamin D deficiency. International organisations such as the World Health Organization (WHO), the Food and Agriculture Organization (FAO), and the United Nations International Children's Fund (UNICEF) have recommended food fortification as one of the most cost-effective and sustainable, population-wide strategies to address micronutrient deficiencies. For example, the WHO and FAO issued the *"Guidelines on Food*

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[1] Vitamin D is a vital micronutrient that plays a crucial role in the development, growth and maintenance of a healthy skeletal system throughout an individual's lifespan. The primary role of Vitamin D is to regulate calcium homeostasis, which is crucial for maintaining proper calcium levels and promoting strong bones. See ICRIER-ANVKA Foundation 2025 report, "*Roadmap to Address Vitamin D Deficiency in India*" (Mukherjee et al., 2025).

[2] Reddy, 2020; Nitish et al., 2024; Bhadada et al., 2021.

[3] https://www.pib.gov.in/PressReleasePage.aspx?PRID=2114715; https://www.pib.gov.in/PressReleasePage.aspx?PRID=2086052; https://www.pib.gov.in/FactsheetDetails.aspx?Id=149135 (Last accessed on July 23, 2025).

[4] https://dhsprogram.com/pubs/pdf/FR375/FR375.pdf (Last accessed on July 16, 2025).

[5] The process of adding one or more vitamins and minerals to commonly consumed food items.

Fortification with Micronutrients" in November 2006, which provides a comprehensive framework for implementing and regulating food fortification programmes.

In India, Food Safety and Standards Authority of India (FSSAI) has permitted the voluntary fortification of milk and edible oil with Vitamin D and mandated its distribution through public food procurement schemes (like Mission POSHAN 2.0).[6]

The Ministry of Food Processing Industries (MoFPI) is the nodal ministry for processing of fortified foods. Under the ministry, institutions such as the National Institute of Food Technology Entrepreneurship and Management (NIFTEM) are already engaged in research and development (R&D) on Vitamin D including bioavailability, stability, formulation, plant-based D_3, Vitamin D retention during cooking, etc. Therefore, the ministry can play a key role in addressing the deficiency through food-based solutions, particularly fortification.

3. Vitamin D Fortification Globally: Experiences of Select Countries

> *"Large scale food fortification is a powerful evidence-informed and cost-effective intervention to fight the consequences of vitamin and mineral deficiencies".*
>
> *-WHO (2023)*

Countries such as Jordan, Nigeria, Sweden, Canada, the United States of America (USA) and Finland have benefited by fortifying a wide range of products such as margarine, milk, cheese, fat spreads, flour, yogurt, juices, biscuits, bread, etc., with Vitamin D. Globally, there are examples of many multi-stakeholder initiatives to address Vitamin D deficiency through food fortification. For example, Bangladesh, in 2011, with the support of the European Union (EU), introduced a school-based micronutrient fortification programme under which daily packets of Vitamin D-fortified biscuits were provided to children in rural primary schools; this has shown positive results in reducing the deficiency (for details see Box 1).

Box 1: Addressing Micronutrient Deficiency through Partnership: The Case of Bangladesh

In 2011, the Bangladesh Government, with support from the European Union (EU), launched a school-based micronutrient food fortification programme. This initiative targeted children in rural primary schools across 10 disadvantaged sub-districts. The intervention provided each child with a daily packet of 75g of biscuits fortified with essential micronutrients, including Vitamin D_3.

The programme had a significant positive impact on increasing mean levels of Vitamin D. Mean serum 25-hydroxyvitamin D [25(OH)D] levels increased by 1.1 times. Fortified biscuits served as an effective vehicle for improving Vitamin D levels, as teachers and caregivers reported improved attentiveness, energy levels and classroom engagement among students.

- Adams et al., (2017)

Studies have shown the positive impact of Vitamin D fortification in reducing the burden of Vitamin D deficiency and of diseases related to the deficiency, as well as improving quality

[6] All such fortified products are required to carry the designated '+F' logo.

adjusted life years (QALYs).[7] For example, in Belgium, Ethgen et al. (2016) evaluated the cost-effectiveness of consuming Vitamin D-fortified yoghurt by 80-year-old women with vertebral fractures and found that it led to a gain of 0.027 QALYs; the incremental cost effectiveness ratio (ICER)[8] was €255 (USD 275) per person. Another study by Aguair et al. (2019) estimated that consuming fortified wheat flour with Vitamin D (400IU per 100 grams) could prevent approximately 9.9 million cases of Vitamin D deficiency and result in the gain of 1.75 million QALYs across the population of England and Wales.

4. Vitamin D Fortification in India – Some Efforts, but are they Enough?

While the Government of India is aware of the rising incidence of Vitamin D deficiency and has taken initiatives, one major miss is in expanding the basket of fortified food products, which countries such as Bangladesh, Pakistan and Nigeria have adopted. This is despite the UNICEF's 2023 guidelines on *"Large-scale Food Fortification for Prevention of Micronutrient Deficiencies in Children, Women and Communities"*, which recommends wheat flour, maize flour and rice as appropriate vehicles for Vitamin D fortification (see Table 1).

Table 1: Potential Food Vehicles and their Fortifiable Micronutrients

Food Vehicle	Vitamins and Minerals (Fortificant) that can be Added
Wheat Flour	Iron, Zinc, Selenium, Vitamins A, D, B_1 (Thiamine), B_2 (Riboflavin), B_3 (Niacin), B_6 (Pyridoxine), B_9 (Folate or Folic Acid), and B_{12} (Cobalamin)
Maize Flour	Iron, Zinc, Vitamins A, D, B_1 (Thiamine), B_2 (Riboflavin), B_3 (Niacin), B_6 (Pyridoxine), B_9 (Folate or Folic Acid), and B_{12} (Cobalamin)
Rice	Iron, Zinc, Selenium, Vitamins A, D, B_1 (Thiamine), B_2 (Riboflavin), B_3 (Niacin), B_6 (Pyridoxine), B_9 (Folate or Folic Acid), and B_{12} (Cobalamin)
Oil	Vitamins A, D, and E
Milk	Vitamins A, D, Iron, and Folic Acid

Source: UNICEF (2023).

Moreover, not everyone in India can afford regular milk consumption. Due to the rising incidence of cardiovascular diseases and obesity, many doctors are now recommending a reduction in the intake of edible oil. Thus, voluntary food fortification of only two products – edible oil and milk – is not enough to prevent the deficiency. Further, a large part of these two products are in the unorganised/informal sector, which do not fortify the products.

Although FSSAI has mandated the use of fortified milk and edible oil in government food procurement programmes (such as Mission POSHAN 2.0), some fortified products cannot be distributed due to supply chain and storage issues. For example, only a few states such as

[7] QALYs measure both the quantity and quality of life lived. One QALY equals one year of life in perfect health. It is calculated by multiplying the number of additional years of life by the health-related quality of life (HRQL) score, which ranges from 0 (equivalent to death) to 1 (perfect health). QALYs are useful for comparing different types of health interventions – for example, one that extends life but with side effects, and another that improves the quality of life without increasing lifespan.

[8] The incremental cost-effectiveness ratio (ICER) is a measure representing the economic value of an intervention, compared with an alternative. In this case, the alternative is no intervention undertaken.
ICER= (Cost of new strategy-cost of current practice)/(Effect of new strategy-effect of current practice).

Maharashtra and Gujarat provide milk through PM POSHAN. Some states like Tamil Nadu, Odisha, West Bengal and Andhra Pradesh have included eggs in their school meals.[9] It will be ideal to distribute fortified milk in partnership with organisations like National Dairy Development Board (NDDB). However, in case there is any issue in the supply chain, then 1 to 2 Vitamin D-enriched biscuits per child may be provided, taking into account that the content should be classified under high fat, sugar and salt food.

Several food products (like packaged juices and cereals) are now available in food stores and non-store retail formats as "Vitamin D Enriched". This is allowed under the FSSAI claim regulation and there is a growing demand for such products. Enriching products with micro-nutrients often makes them a premium product, with the high price reducing their affordability to the masses.

5. Action Points for MoFPI to Address Vitamin D Deficiency in India

The following five action points by the MoFPI will help to address Vitamin D deficiency in India.

5.1 Collaborate with FSSAI and other Stakeholders for more Vitamin D Fortified Food Products

To realise the vision of *"Vitamin D Kuposhan Mukt Bharat***"**, the FSSAI Scientific Panel on Food Fortification, experts from the NIFTEM, Central Food Technological Research Institute (CFTRI), and other stakeholders [such as the Global Alliance for Improved Nutrition (GAIN) and the Gates Foundation] may work together to achieve the following:

- **Expand the Range of Fortifiable Food Vehicles:** At present, fortification efforts are limited to edible oils and milk. Identifying additional food vehicles suitable for Vitamin D fortification, such as commonly consumed staples like rice, wheat flour and maize flour can significantly help address Vitamin D deficiency. In this regard, UNICEF's 2023 *Guidelines on Large-Scale Food Fortification* (see Table 1) may be examined. In India, various organisations are already engaged in developing Vitamin D-fortified products like biscuits. These can be distributed to at-risk populations under public nutrition programmes such as Mission POSHAN 2.0.

- **Allow Fortification with Both Vitamin D_2 and D_3:** Currently, Vitamin D fortification in India is limited to plant-based forms of Vitamin D. Expanding the scope to include animal-based Vitamin D – with appropriate non-vegetarian labelling – would allow consumers to make

[9] https://dsel.education.gov.in/sites/default/files/schemes_guidelines/Guidelines_PM%20POSHAN_SCHEM E.pdf; https://www.niti.gov.in/sites/default/files/2022-06/Take-home-ration-report-30_06_2022.pdf (Last accessed on July 29, 2025).

informed choices based on their dietary preferences. While NIFTEM and others are conducting R&D on Vitamin D_3 from plant sources, industry buy-in and the engagement of healthcare professionals are needed to help in the commercialisation of such products and to spread knowledge and reduce misconceptions about Vitamin D from plant and animal-based sources.

- **Create a Multi-stakeholder Platform to Discuss and Share Research on Bioavailability and Stability of Vitamin D Forms:** NIFTEM is conducting analysis on the bioavailability, stability and absorption rates of different forms of Vitamin D (e.g., D_2 vs. D_3) in various food matrices. However, the food processing industry is often concerned about the potential loss of intellectual property (IP) during the research process; strengthening IP protection further is, therefore, essential. A well-designed multi-stakeholder platform can facilitate collaboration and serve as a forum for discussing and sharing research in a secure and structured manner.
- **Work with ICMR-NIN and FSSAI to Make Vitamin D Fortified Food a part of Dietary Guidelines 2024:** The 2024 *Dietary Guidelines* by ICMR-NIN highlight sunlight as a key source of Vitamin D but does not mention food as a vehicle to reduce Vitamin D deficiency. MoFPI may work with ICMR-NIN and FSSAI to include Vitamin D rich and fortified foods as part of the addendum to 2025 *Dietary Guidelines*. While it will help the consumer adopt a rich diet, it will also help the industry adopt fortification during food processing.
- **Work with Industry to Build Consensus on Mandatory Fortification:** It is necessary to engage with industry to understand their concerns (such as cost implications, reformulation challenges, compliance burden, etc.) and provide targeted support to help shift from voluntary to mandatory fortification.

5.2 Support Research and Innovation in Vitamin D Fortification

Targeted research and development are essential to support the advancement of Vitamin D fortification in India. NIFTEM, in collaboration with companies like Fermenta Biotech, are already working towards developing plant-based sources of Vitamin D. Such collaborations may be encouraged. Other areas of research may include the following:

- **Preserving Vitamin D during Cooking:** Vitamin D is relatively stable during cooking; however, repeated boiling and exposure to high temperatures, such as those used in slow cooking and frying, can reduce the content of Vitamin D in food. This underscores the need for further research on the retention of Vitamin D after frying, particularly in the context of Indian culinary practices.
- **Exploring Innovative Fortification Techniques:** Techniques such as ultraviolet (UV) exposure of mushrooms to increase their Vitamin D content are already underway. These innovations should be studied further to assess how they can be scaled and applied to other food processing methods.

- **Develop more Vitamin D Fortified Products:** There is a need to develop new Vitamin D-fortified products, (such as ready-to-eat snacks, protein bars and bakery items) to increase options for the consumer. All newly developed fortified products must undergo rigorous testing and food trials to ensure they are safe, effective and acceptable to consumers.
- **Comparative Analysis of Vitamin D_2 and D_3:** R&D can be supported to compare the effectiveness, bioavailability and cost-efficiency of Vitamin D_2 and D_3.

  The above areas of research require sustained investment and institutional support through targeted funding, innovation grants and public-private partnerships. MoFPI can play a key role in supporting a nationwide *"Vitamin D Kuposhan Mukt Bharat"* campaign by integrating these research priorities and collaborating with the Ministry of Finance to secure dedicated funding for the campaign.

5.3 Launching Targeted Pilot Programmes through Multi-Stakeholder Partnerships

The MoFPI may collaborate with ICMR-NIN (MoHFW), Ministry of Women and Child Development (MoWCD) and organisations such as GAIN, the Gates Foundation, ANVKA Foundation, etc., to launch pilot food fortification projects in high-risk districts and among vulnerable population groups. The ICMR-NIN Survey for Assessment of Markers of Population Health Activity, Diet and Anthropometry (SAMPADA)[10] can be used to identify the targeted districts and vulnerable groups for pilots and fortified foods like biscuits can be given to check efficacy, ensuring that portions are controlled.

5.4 Incentivising Fortified Food Production and Consumption

The MoFPI can work with the Ministry of Finance to promote the production of Vitamin D fortified food by providing nutrition linked incentives to manufacturers for mandatory fortification and for R&D in Vitamin D fortified products.

5.5 Capacity Building of Micro, Small and Medium Enterprises (MSMEs) for Vitamin D Fortification

The MoFPI can strengthen Vitamin D fortification efforts by building the capacity of MSMEs. This can be achieved through targeted training programmes, technical support and simplified compliance guidance, enabling MSMEs to adopt fortification practices effectively. Organisations such as the GAIN can support these efforts by providing toolkits, technological assistance and best practices for cost-effective and safe food fortification.

To conclude, NIFTEM under MoFPI is actively working on food fortification with essential micronutrients, including Vitamin D. The MoFPI is well-connected with the food industry and by embedding nutrition in food processing and working with other government agencies and stakeholders, they can make fortified foods reach the most vulnerable, MoFPI can play a transformative role in realising a *"Vitamin D Kuposhan Mukt Bharat".*

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[10] SAMPADA is India's first diet and bio-marker survey covering 36 States and Union Territories (UTs), with a sample size of 2.5 lakh individuals and a sub-sample of 30,000 specifically tested for micronutrient deficiency, including Vitamin D.

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ICRIER has two office locations in Delhi; in the institutional complex of India Habitat Centre and a new office at the Institutional Area, Sector 6, Pushp Vihar, New Delhi.

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ANVKA
Foundation

The ANVKA Foundation, a Section 8 registered non-governmental organization, is committed to fostering holistic community development and social transformation. Led by renowned surgeon and philanthropist Dr. Aashish Chaudhry, the foundation's initiatives span education, social welfare, cultural preservation, economic empowerment, medical relief and environmental protection, all aimed at uplifting underprivileged communities. Rooted in practical philosophy, art, aesthetics and psychoanalytic insights, ANVKA addresses critical societal issues such as education, gender equality, ecology, governance and public health, with a focus on both rural and urban areas. Collaborating with government and community-driven initiatives, ANVKA ensures comprehensive support across diverse needs.

At the heart of ANVKA's mission is the promotion of innovative education that integrates the socio-political, economic, spiritual and aesthetic dimensions. The foundation advocates the re-education of relationships, challenging patriarchal norms and fostering mutual respect and dignity to build a more equitable society. ANVKA works to dismantle oppressive social structures, focusing on creating a future grounded in love, understanding and equality. Additionally, the foundation promotes environmental sustainability and livelihoods that are ecologically responsible and socially fair, recognizing the interconnectedness of all life and striving for harmony between humans and the environment. Their efforts also include promoting active citizenship by encouraging individuals to engage in participatory democracy and take an active role in shaping their communities.

ANVKA's health interventions prioritize preventive care for vulnerable populations, emphasizing the importance of building a culture of compassionate community support. Through these initiatives, the foundation empowers communities to actively participate in shaping their futures, fostering social justice and promoting inclusivity. ANVKA's core areas of focus include education, gender equality, empowerment, environmental protection, community development, and health and well-being. Their innovative educational programs are grounded in lived experiences, integrating multiple dimensions, while their gender equality initiatives challenge societal norms to build respectful, equitable relationships. The foundation promotes livelihoods that are economically viable and environmentally sustainable, while also advocating for a harmonious relationship with nature.

Guided by values of holistic well-being, community engagement, social justice, sustainability and innovation, ANVKA strives to build a just, inclusive and thriving world for all. Together, we can create a brighter, more equitable future.

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